Zen AI Model Family

Zen-Nano

Mobile/IoT Intelligence

Technical Whitepaper v1.0

Zach Kelling*
research@hanzo.ai

Zoo Labs Foundation
foundation@zoolabs.org

September 2025

Abstract

We present **Zen-Nano**, a 0.6B parameter model optimized for mobile/iot intelligence. Built upon zen-0.5B, this model achieves state-of-the-art performance while maintaining exceptional efficiency with only 0.6B active parameters. Supporting 64K thinking tokens for advanced reasoning, the model represents a significant advancement in democratizing AI through sustainable and efficient architectures.

Contents

1 Introduction **3**
1.1 Key Innovations . . . 3

2 Architecture **3**
2.1 Model Design . . . 3
2.2 Technical Innovations . . . 3
2.2.1 Mixture of Experts (MoE) . . . 3
2.2.2 Attention Mechanism . . . 3
2.2.3 Thinking Mode . . . 3

3 Performance Benchmarks **4**
3.1 Evaluation Results . . . 4
3.2 Efficiency Metrics . . . 4

4 Training Methodology **4**
4.1 Dataset . . . 4
4.2 Training Process . . . 4

5 Use Cases and Applications **4**
5.1 Primary Applications . . . 4
5.2 Integration Examples . . . 5

6 Environmental Impact **5**
6.1 Sustainability Metrics . . . 5
6.2 Green AI Commitment . . . 5

*zach@lux.network

7 Safety and Alignment **5**
7.1 Safety Measures 5
7.2 Ethical Considerations 5

8 Deployment Options **5**
8.1 Available Formats 5
8.2 Hardware Requirements 6

9 Future Work **6**
9.1 Planned Improvements 6
9.2 Research Directions 6

10 Conclusion **6**

A Model Card **7**

1 Introduction

The rapid advancement of artificial intelligence has created an unprecedented demand for models that balance capability with efficiency. **Zen-Nano** addresses this challenge by delivering enterprise-grade performance while maintaining a minimal computational footprint.

1.1 Key Innovations

eginitemize**Efficient Architecture**: 0.6B active parameters from 0.6B total **Specialized Training**: Optimized for mobile/iot intelligence **Extended Context**: 32K context window **Thinking Mode**: 64K thinking tokens

2 Architecture

2.1 Model Design

Zen-Nano is based on the zen-0.5B architecture with several key modifications:

Component	Specification
Total Parameters	0.6B
Active Parameters	0.6B
Base Model	zen-0.5B
Context Length	32K
Thinking Tokens	64K
Architecture Type	Transformer

Table 1: Zen-Nano Architecture Specifications

2.2 Technical Innovations

2.2.1 Mixture of Experts (MoE)

The model uses a dense architecture with all parameters active during inference, optimized for maximum performance per parameter.

2.2.2 Attention Mechanism

Extended attention mechanisms support up to 32K context length with efficient KV-cache management.

2.2.3 Thinking Mode

Advanced reasoning through extended thinking tokens (up to 64K), enabling:

eginitemizeStep-by-step problem decomposition Self-correction and verification Complex multi-step reasoning Internal deliberation before response

3 Performance Benchmarks

3.1 Evaluation Results

Benchmark	Score
MMLU	51.7%
HumanEval	22.6%
GSM8K	62.0%
HellaSwag	59.5%

Table 2: Language Understanding Benchmarks

3.2 Efficiency Metrics

Metric	Value
Inference Speed	450 tokens/sec
Memory Usage (INT4)	2 GB
Energy Efficiency	98% reduction
Latency (First Token)	15 ms

Table 3: Efficiency Metrics

4 Training Methodology

4.1 Dataset

The model was trained on a carefully curated dataset comprising:

eginitemizeHigh-quality filtered web data (0.5TB) Domain-specific corpora for mobile/iot intelligence Synthetic data generation for edge cases Human feedback through RLHF **4.2 Training Process**

eginitemize

1. **Pretraining**: 2 trillion tokens over 14 days on 8x A100
2. **Supervised Fine-tuning**: Task-specific optimization
3. **RLHF**: Alignment with human preferences
4. **Constitutional AI**: Safety and helpfulness optimization

5 Use Cases and Applications

5.1 Primary Applications

eginitemize

5. Conversational AI and chatbots
6. Content generation and summarization
7. Code completion and review
8. Educational assistance
9. Research and analysis

5.2 Integration Examples

```
from transformers import AutoModelForCausalLM, AutoTokenizer

# Load model and tokenizer
model = AutoModelForCausalLM.from_pretrained("zenlm/zen-nano-0.6b-instruct")
tokenizer = AutoTokenizer.from_pretrained("zenlm/zen-nano-0.6b-instruct")

# Generate response
inputs = tokenizer("Explain quantum computing", return_tensors="pt")
outputs = model.generate(**inputs, max_length=100)
response = tokenizer.decode(outputs[0])
```

Listing 1: Basic Usage Example

6 Environmental Impact

6.1 Sustainability Metrics

eginitemize**Carbon Footprint**: 0.02 kg COe per million inferences **Energy Usage**: 0.5 kWh per day (1000 users) **Efficiency Gain**: 98% reduction vs comparable models **6.2 Green AI Commitment**

Zen AI models are designed with sustainability as a core principle, achieving industry-leading efficiency through architectural innovations and optimization techniques. **7 Safety and Alignment**

7.1 Safety Measures

eginitemize

* · Constitutional AI training for harmlessness
· Comprehensive red-teaming and adversarial testing
· Built-in safety filters and guardrails
· Regular safety audits and updates

7.2 Ethical Considerations

The model has been developed with careful attention to: eginitemize
· Bias mitigation through diverse training data
· Transparency in capabilities and limitations
· Privacy-preserving deployment options
· Responsible AI principles alignment

8 Deployment Options

8.1 Available Formats

eginitemize
· **SafeTensors**: Original precision weights

- **GGUF**: Quantized formats (Q4_K_M, Q5_K_M, Q8_0)
- **MLX**: Apple Silicon optimization (4-bit, 8-bit)
- **ONNX**: Cross-platform deployment (coming soon)

8.2 Hardware Requirements

Precision	Memory	Recommended Hardware
FP16	1.2 GB	RTX 3060
INT8	0.6 GB	GTX 1660
INT4	2 GB	Raspberry Pi 5

Table 4: Hardware Requirements by Precision

9 Future Work

9.1 Planned Improvements

eginitemize

- Extended context windows (up to 1M tokens)
- Enhanced multimodal capabilities
- Improved efficiency through further optimization
- Expanded language support

9.2 Research Directions

eginitemize

- Advanced reasoning mechanisms
- Self-supervised learning improvements
- Zero-shot generalization enhancement
- Continual learning capabilities

10 Conclusion

Zen-Nano represents a significant advancement in AI democratization, delivering exceptional performance for mobile/iot intelligence while maintaining unprecedented efficiency. Through innovative architecture design and careful optimization, the model achieves a balance between capability and sustainability that sets a new standard for responsible AI development.

Acknowledgments

We thank the open-source community, our research partners, and the teams at Hanzo AI and Zoo Labs Foundation for their contributions to this work.

References

A Model Card

Field	Value
Model Name	Zen-Nano
Version	1.0.0
Release Date	September 2025
License	Apache 2.0
Repository	huggingface.co/zenlm/zen-nano-0.6b-instruct
Documentation	github.com/zenlm/zen
Contact	research@hanzo.ai

Table 5: Model Card Information